Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 3103
(Materials Manufacturing Processes)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Why is casting process considered as a major manufacturing process? Discuss the disadvantages of the casting process? (08)

   (b) How do you create quality casting? What are the limitations of casting? (15)

   (c) Why is homogeneous nucleation of liquid metal difficult? How do foreign particles help in nucleating solid particles in liquid metal? What kind of foreign particles are most suitable for this job? (12)

2. (a) Explain major casting defects and their reasons? (15)

   (b) Write short notes on the mechanisms of clay-water bonding in green molding sand aggregates. (10)

   (c) Briefly describe the steps of sand mold casting. (10)

3. (a) What is undercooling? Examine how the depression of undercooling influence the solidified structure of pure metals and alloys. (13)

   (b) Describe the mechanism of equiaxed crystals formation. (10)

   (c) Briefly describe any three feeding rules. (12)

4. (a) Analyze the characteristics difference of hot tear, shrinkage cavity and cold crack. Discuss how you can reduce or eliminate hot tear defects. (10)

   (b) A thin-wall cylindrical casting of 60 mm diameter (inner), 8 mm thickness and 170 mm length is to be cast using cast iron in greensand mould. Determine the followings- (25)

   i) Using 20% longer than solidification time, calculate the pouring time required for the casting. The mould constant in Chvorinov's rule is 1.85 s/mm^2.
   ii) Using 50 mm high-pouring basin, calculate the effective metal head pressure developed in the gating system. List all assumptions you have made.
   (iii) Calculate the choke area of the gating system. Use 7.0 g/cc for liquid metal density and 0.90 for discharge coefficient.
   (iv) Calculate sprue top and bottom areas, runner area and gate area of the gating system. Use critical velocity = 500 mm/s for this gating system with suitable gating ratio. Also use a safety factor of 30% to oversize the sprue top area.

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Describe the required properties of polymer melt. (10)

   (b) Which one is technically more feasible and economical for polymer double-layer bottles making between extrusion blow molding and injection blow molding? Justify your answer. (15)

   (c) The density and associated percent crystallinity for two poly (ethylene terephthalate) materials are as follows: (10)

ρ (g/cc)	Crystallinity (%)
1.452	78.7
1.375	35.3

   Compute the densities of totally crystalline and totally amorphous poly (ethylene terephthalate).

6. (a) Write short notes on (i) Drilling operation and (ii) Boring operation (10)

   (b) What types of defects are observed in polymer extruded products? Describe the causes and remedies of one of these. (12)

   (c) Briefly describe the steps required for a powder metallurgy process. (13)

7. (a) How does the Sintering operation reduce porosity and improve mechanical strength of ceramic powder products? (08)

   (b) Briefly discuss the cementation process of Portland cement with necessary chemical reactions. (13)

   (c) Which AM technique will you prefer for photosensitive liquid polymer fabrication? Briefly describe this AM technique. (14)

8. (a) What is the advantages of machining? What properties should cutting tools have? (06)

   (b) What do you understand by ~~steel~~ sheet metal forming? Describe the defects and remedy in sheet metal forming. (12)

   (c) A cylindrical workpiece is subjected to a cold, upset forging operation. The starting piece is 75 mm in height and 50 mm in diameter. It is reduced in the operation to a height of 36 mm. The work material has a flow curve defined by K=350 MPa and n=0.17. Assume a coefficient of friction of 0.1. Determine the force as the process begins and at the final height of 36 mm. (07)

   (d) Which additive manufacturing route is the best for metallic element production? Justify your answer. (10)

Khulna University of Engineering & Technology

B.Sc. Engineering 3rd Year 1st Term Examination, 2021

Department of Materials Science and Engineering

MSE 3105

(Welding and Materials Joining Processes)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) Describe Electroslag welding process with its advantages and disadvantages. (14)

(b) What are the differences between soldering and brazing. (08)

(c) How does molten metal bath dip brazing differ from molten chemical (flux) bath dip brazing? (13)

Q 2. (a) With appropriate diagrams, explain different types of weld joints. (10)

(b) Write down the characteristics of Brazing fluxes. (10)

(c) Suppose, you are given with two samples: Aluminium and steel (Incompatible metal). For the joining of those two dissimilar metals which type of friction welding process would you prefer? And why? Then, also explain its working principle. (15)

Q 3. (a) Name the time and temperature dependent solid state welding process and hence, describe the mechanism of this type of welding process with necessary diagram. (15)

(b) What are the functions of electrode covering in SMAW? Discuss briefly. (10)

(c) What are the functions of shielding gases in gas-metal arc welding? (10)

Q 4. (a) Which current mode would you choose for welding thin sheet in GTAW process? Between DCEN and DCEP mode which one produces cleaner surface and how? (07)

(b) Draw the diagram of desired welds from the weld symbols shown in figure 4(b). (10)

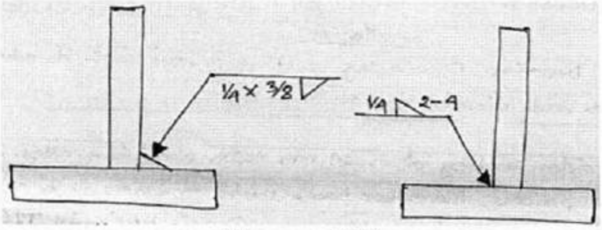

Figure: 4(b)

(c) Describe one of the High-Energy-Density Beam welding. (10)

(d) How does friction welding differ from diffusion welding? (08)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Sketch a weld pool microstructure. (05)

(b) Distinguish between epitaxial growth and non-epitaxial growth at the fusion boundary. (10)

(c) How does grain structure influence the mechanical properties of weld? (10)

(d) An Al-1% Cu alloy is welded autogenously by GTAW and an Al-5% Cu alloy is welded under identical condition. Which alloy is likely to have more equiaxed dendrites in the weld metal and why? (10)

Q 6. (a) Explain a well-known constitutional liquation mechanism with the help of the phase diagram and microstructure given in figure 6(a). (13)

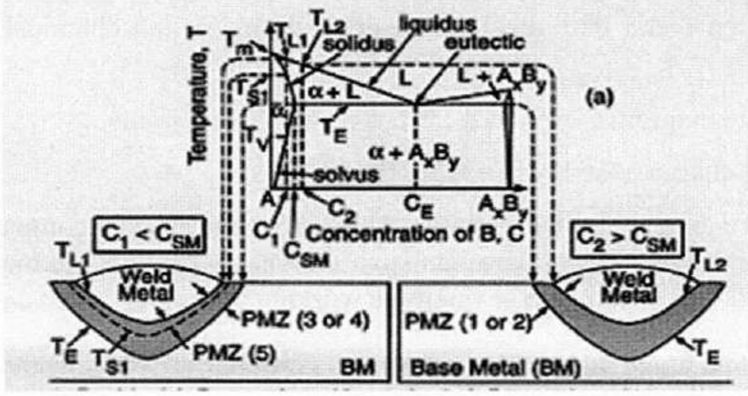

Figure: 6(a)

(b) Briefly describe the phenomenon of liquation cracking. (12)

(c) Briefly discuss the different types of rivet joints. (10)

Q 7. (a) Briefly describe the phenomena of weld decay in HAZ. What are the remedies for weld decay? (13)

(b) Which joining process will you prefer for joining the metal-ceramic system? Justify your answer. (10)

(c) What is knife line attack? How can this problem be reduced? (12)

Q 8. (a) What are the factors that affect the crack susceptibility? Explain. (10)

(b) What is NDT? How can the subsurface defects in a weldment of ferromagnetic material be identified by NDT? Explain this testing method. (15)

(c) What is angular distortion in a weldment? Why does it form? Discuss any two remedies for angular distortion. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 3107
(Physical Metallurgy of Materials)

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Define cold working process. Explain the recrystalline mechanism of a cold worked sample with necessary diagram. (10)

   (b) Briefly discuss the factors that affect the recrystallied grain size. (15)

   (c) What type of changes in mechanical properties and micro-structure would you observe in a sample which is just stress relieved? Explain with necessary sketches. (10)

2. (a) Draw a neat sketch of iron-iron carbide diagram with proper labeling and illustrate the microstructural changes during cooling from liquid state of the following samples with: (i) 4.3 % Carbon (ii) 1.5 % Carbon (iii) 0.5 % Carbon composition. (15)

   (b) What are the effects of Nickel, Molybdenum, Phosphorus and Silicon on the properties of steel? (10)

   (c) Define CCT diagram. Discuss the effects of alloying elements on the position of I-T diagram. (10)

3. (a) Define 'texture' and explain why it is important to control the texture of an engineering material? (05)

   (b) What are the criteria for superalloys? Write down the major phases in Ni-based superalloys. Among these phases, which one do you think to be harmful? Explain. (15)

   (c) "Austempering is an isothermal process"-justify the statement with TTT diagram. (15)

4. (a) Describe the heat treatment process of a UHSS with necessary diagram. (10)

   (b) What is tempering of steel? Shortly discuss the different types of tempering process. (10)

(c) Draw an I-T diagram and sketch the cooling curves to obtain (i) 100% bainite (ii) 30% pearlite, 30% bainite and the remaining martensite (iii) 50% fine pearlite and 50% martensite. (15)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) How does white nitride layer form in low Carbon steel? Explain how it affects the mechanical properties of steels? (10)

   (b) Compare and contrast between carbonitriding and cyaniding? (10)

   (c) Briefly discuss the gas carbonizing process with necessary figures. (15)

6. (a) Write down some names of surface hardening process that need further heat treatment. Also, explain the reason. (10)

   (b) What are the limitations of Austempering and Martempering processes? (10)

   (c) Briefly explain the nitriding process. Illustrate the microstructural changes of steel in nitriding operation. (15)

7. (a) What are the purpose of heat treatment of gray cast iron? (10)

   (b) What are the important criteria to be considered to select ageing temperature for precipitation hardening? Explain with example. (10)

   (c) What is meant by malleablization of white cast iron? With a neat sketch, discuss the heat treatment of white cast iron. (15)

8. (a) Write short notes on: (i) Super alloy (ii) Micro-alloyed steel (10)

   (b) What is TMT alloy? How can you get high toughness and hardness from TMT alloy? Explain with necessary figures. (12)

   (c) Briefly discuss the dispersion strengthening mechanism with appropriate example. (13)

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2021
Department of Materials Science and Engineering

Hum 3127
(Government and Sociology)

Time: 3hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) What do you mean by politics? Is politics essential for the development of a country? Why? (15)

   (b) What is state and government? What are the functions of a modern welfare state? (20)

2. (a) What is the relationship between the presidential and parliamentary form of government? Is a presidential system more democratic than a parliamentary system? Why or why not? (20)

   (b) What is citizenship? What are the qualities of an ideal citizen? What are the hindrances in the way of ideal citizen? (15)

3. (a) Critically discuss the theory of divine origin of the state? (10)

   (b) What are the main features of Marxism? What were Marx's main contributions for understanding society and struggle of liberation? (15)

   (c) Distinguish between the compulsory and optional function of a modern welfare state. (10)

4. (a) What is constitution? Do you think that the constitution is of Bangladesh is a good constitution in the world and why? (10)

   (b) What is fascism? Give an account of Mussolini's fascism in detail. Discuss the problems that fascism in Italy could not solve. (15)

   (c) What are the Legislature, Executive and Judiciary? What are the functions of Judiciary? (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Who is the founding father of sociology? Explain sociology from his point of view? (05)

   (b) Write down the contribution of Herbert Spencer to the development of sociology. (15)

   (c) What are the scopes of sociology? Explain in brief. (15)

6. (a) Which factors dissociate community from society? (10)

   (b) What are the major elements of social structure? Explain it with example. (10)

   (c) "The forms of the social stratification's are universal"- discuss it from your own perspective. (15)

7. (a) What is the meaning of culture? When does cultural lag occur in a society? (15)

   (b) Discuss the role of family as an agent of socialization for a children. (10)

   (c) Critically evaluate Cooley's theory of "Looking-Glass-Self". (10)

8. (a) Define social change. Explain the different forms of social change. (10)

   (b) What is Ethnocentrism? Is Ethnocentrism good or bad? (10)

   (c) Explain the different stages of socialization with example. (15)

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 3101
Transport Phenomena in Materials

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) Write down the three major assumptions used in the derivation of the Bernoulli equations and derive the Bernoulli equation. (20)

(b) A piezometer and a Pitot tube are tapped into a horizontal water pipe, as shown in figure 1(b) to measure static and stagnation (static + dynamic) pressures. For the indicated water column heights, determine the velocity at the center of the pipe. (10)

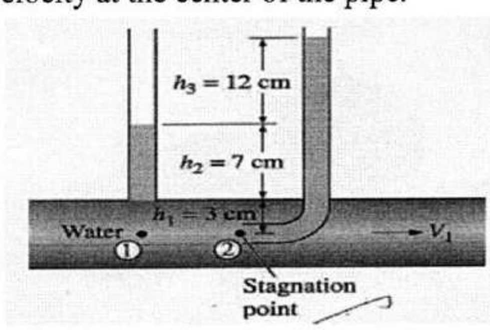

Figure: 1(b)

(c) List the limitations on the use of Bernoulli Equation. (05)

Q 2. (a) Consider a slab of thickness L and constant thermal conductivity k in which energy is generated at a constant rate of $g_\circ w/m^3$. The boundary surface at x = 0 is insulated and that at x = L dissipates heat by convection with a heat transfer co-efficient h into a fluid at a temperature T_∞. Develop expressions for the temperature T(x) and the heat flux q(x) in the slab. (15)

(b) A 60cm wide belt moves as shown in fig. 2(b). Calculate the horsepower requirement assuming a linear velocity profile in the 10°C water. μ for 10°C water = 1.38×10^{-3} N.s/m^2 (10)

(c) Suppose that the fluid being sheared in fig. 2(c) is SAE 30 oil at 20°C. Compute the shear stress in the oil if V = 3 m/s and h = 2 cm. For SAE 30 oil, viscosity, $\mu = 0.29\ kg/m$-s. (10)

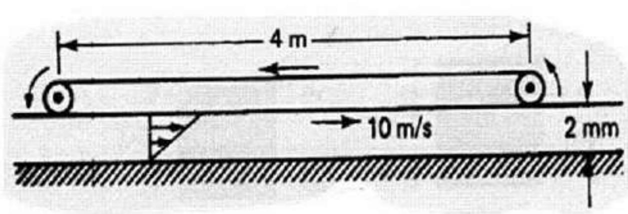

Figure: 2(b)

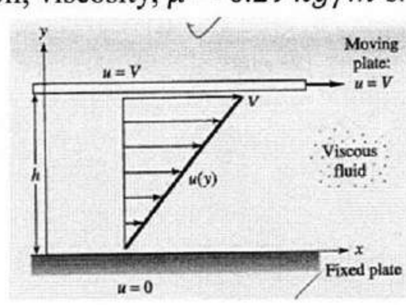

Figure: 2(c)

Q 3. (a) Consider a stem pipe of length L= 20 m, inner radius $r_1 = 6$ cm, outer radius $r_2 = 8$ cm and thermal conductivity $k = 20\ W/m.°C$, as shown in fig. 3(a). The inner and outer surfaces of the pipe are maintained at average temperatures of $T_1 = 150°C$ and $T_2 = 60°C$, respectively. Obtain a general relation for the temperature distribution inside the pipe under steady conditions and determine the rate of heat loss from the steam through the pipe. (18)

(b) A steel tube with 5 cm inner diameter, 7.6 cm outer diameter and $k = 15\ W/m.°C$ is covered with an insulative covering of thickness t = 2 cm and $k = 0.2\ W/m.°C$. A hot gas at $T_a = 330°C$, $h_a = 400\ W/m^2.°C$ flows inside the tube. The outer surface of the insulation is exposed to cooler air at $T_b = 30°C$, with $h_b = 60\ W/m^2.°C$. Calculate the heat loss from the tube to the air for length L = 10 m of the tube. Calculate the temperature drops resulting from the thermal resistances of the hot gas flow, the steel tube, the insulation layer and the outside air. (17)

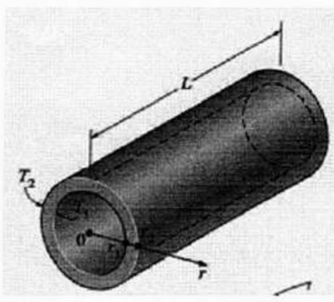

Figure: 3(a)

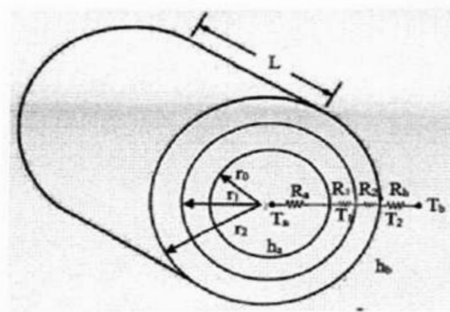

Figure: 3(b)

Q 4. (a) An iron sphere [$k = 60\ W/m.°C$, $C_p = 460\ J/kg.°C$, $\rho = 7850\ kg/m^3$, and $\alpha = 1.6 \times 10^{-5}\ m^2/s$] of diameter D = 5 cm is initially at a uniform temperature $T_i = 225°C$. Suddenly the surface of the sphere is exposed to an ambient at $T_\alpha = 25°C$ with a heat transfer coefficient $h = 500 W/m^2.°C$. (18)

(i) Calculate the center temperature t = 2 min after the start of the cooling.

(ii) Calculate the temperature at a depth 1cm from the surface t = 2 min after the start of the cooling.

(iii) Calculate the energy removed from the sphere during this time period.

[Use the charts in figure 4(a) if necessary]

(b) A 3 mm diameter and 5 m long electric wire is tightly wrapped with a 2 mm thick plastic cover whose thermal conductivity is $k = 0.15\ W/m.°C$. Electrical measurements indicate that a current of 10 A passes through the wire and there is a voltage drop of 8 V along the wire. If the insulated wire is exposed to a medium at $T_\alpha = 30°C$ with a heat transfer coefficient of $h = 12\ W/m^2.°C$, determine the temperature at the interface of the wire and the plastic cover in steady operation. Also determine whether doubling the thickness of the plastic cover will increase or decrease this interface temperature. (17)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) With a neat sketch, explain laminar and turbulent velocity boundary layers that may form on a flat plate. (15)

(b) Air at 27°C and 1 atm flows over a flat plate at a speed of 2 m/s. Calculate the boundary layer thickness at distances of 20 cm and 40 cm from the leading edge of the plate. The velocity of air at 27°C is 1.85×10^{-5} kg/m.s. Assume unit depth in the Z direction. (12)

(c) What is the physical significance of the Reynolds number? (08)

Q 6. (a) Derive an expression for the log mean temperature difference (LMTD) for a counter flow heat exchanger. (18)

(b) A counter flow double pipe heat exchanger is to heat water from 20°C to 80°C at a rate of 1.2kg/s. The heating is to be accomplished by geothermal water available at 160°C at a mass flow rate of 2 kg/s. The inner tube is thin-walled and has a diameter of 1.5 cm. If the overall heat transfer coefficient of the heat exchanger is 640 $W/m^2.°C$, determine the length of the heat exchanger required to achieve the desired heating. (17)

Q 7. (a) Prove that the solid angle associated with the entire hemisphere is 2π sr. (15)

(b) A small surface of area $A_1 = 10^{-3}m^2$ is known to emit diffusely and from measurements the total intensity associated with emission in the normal direction is $I_n = 7000\ W/m^2$ sr. Radiation emitted from the surface is intercepted by three other surfaces of Area $A_2 = A_3 = A_4 = 10^{-3}m^2$. Which are 0.5 m from A_1 [fig. 7(b)]. (15)

i) What are the solid angles subtended by the three surfaces when viewed from A_1?

ii) What is the rate at which radiation emitted by A_1 is intercepted by the three surfaces?

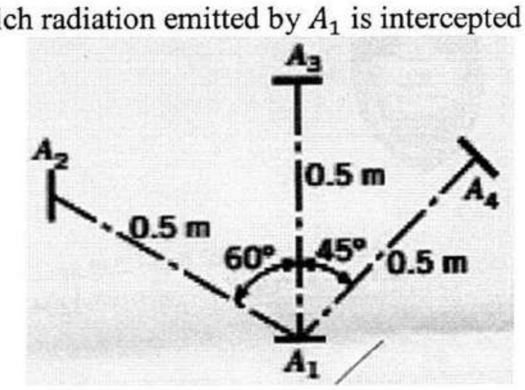

Figure: 7(b)

(c) What is Wien's displacement law? (05)

Q 8. (a) For equal emissivities, prove that radiation exchange between large parallel plane with a radiation shield is half of the radiation exchange without a shield. (15)

(b) Consider the diffusion of hydrogen in air at T = 293K. Calculate the species flux on both molar and mass bases the concentration gradient $^{dc_A}/_{dx} = 1\ Kmol/m^3.m$. Here, $D_{AB} = 0.41 \times 10^{-4} m^2/s$ at 298K. (10)

(c) Determine the view factors F_{12} and F_{21} for the following geometries. (10)

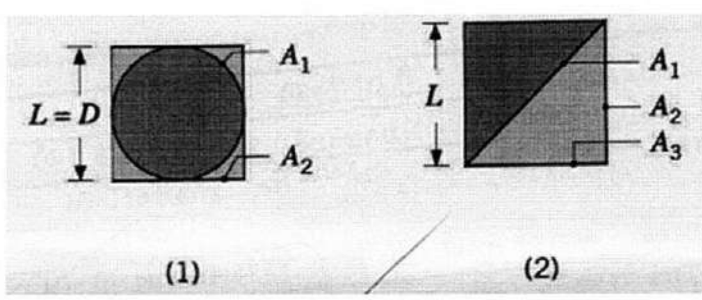

Figure: 8(c)

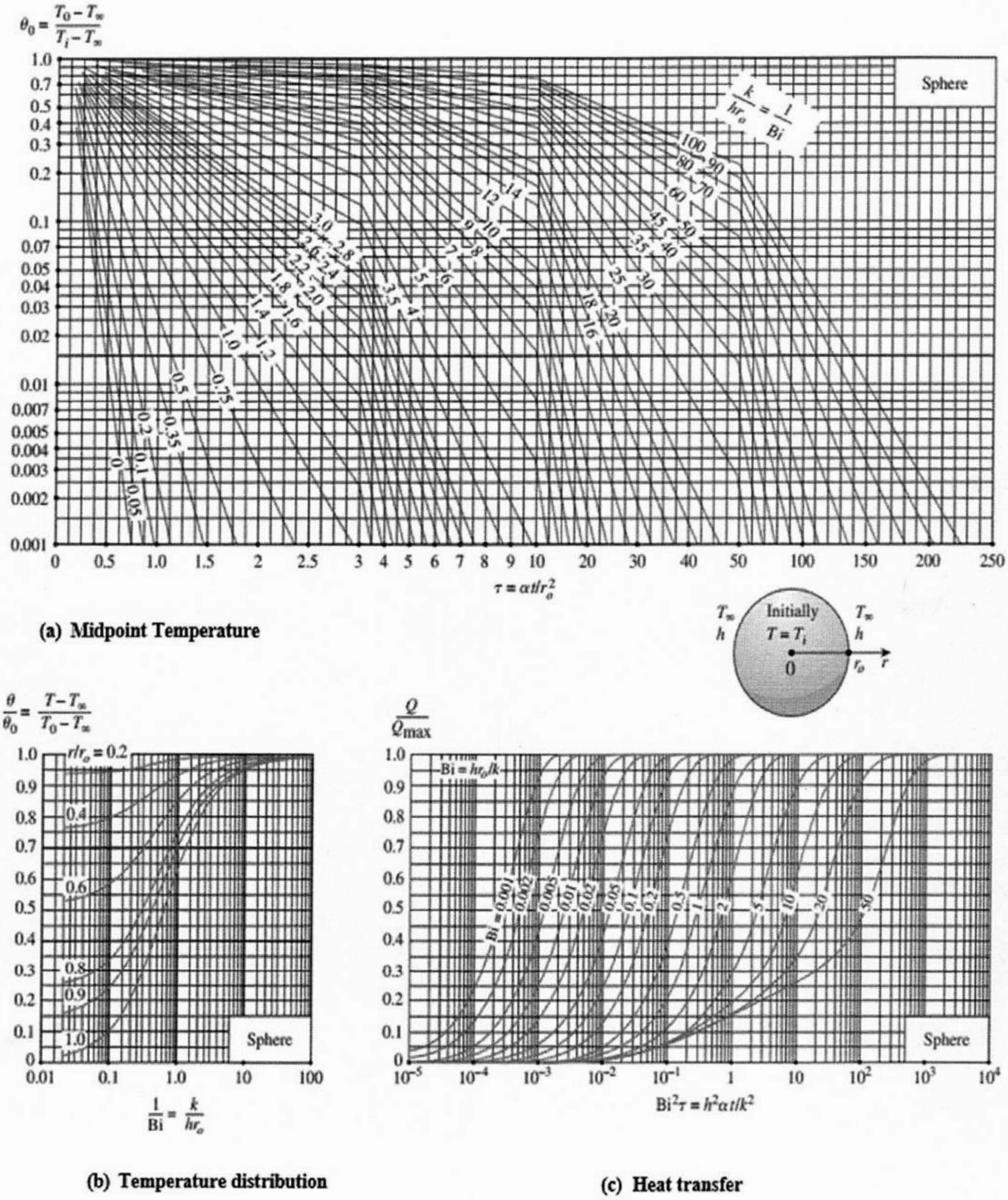

Figure 4(a) Transient temperature and heat transfer charts for a sphere of radius r_o initially at a uniform temperature T_i subjected to convection from all sides to an environment at temperature T_∞ with a convection coefficient of h.